SE4041 · MOBILE APPLICATION DESIGN AND DEVELOPMENT

GREENMIND: Plant Care Dashboard & Smart Garden Ecosystem

Assignment 02 Comprehensive Technical Documentation · Semester 2, 2026

25% TOTAL
Part A (15%) + Part B (10%)

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Degree Programme	BSc (Hons) in Information Technology	**Year & Semester**	Year 4, Semester 2
Document Type	Working Architecture & Audit Draft	**Target Deadline**	November 2026

1. Assignment Overview & Executive Summary

GreenMind is a digital plant-care management companion engineered to organize domestic plant collections and present glanceable condition indicators. Across dual platforms, GreenMind addresses domestic plant mortality caused by inconsistent irrigation: **Part A** delivers an advanced iOS client featuring telemetry and persistent stores, while **Part B** translates the experience into a specialized living-room tvOS dashboard using remote-driven navigation.

> **Academic Audit Note:** This document reflects evidenced code, active prototypes, and planned components. Features marked *Planned* or *Verification Required* are architectural commitments and must be updated with final production test logs prior to formal viva submission.

2. Part A: Advanced iOS Application Scope (15%)

The brief mandates a fully functional iOS app built with declarative SwiftUI, robust navigation, custom components, persistence, and an emerging Apple framework.

Brief Requirement	GreenMind Architectural Alignment	Status / Verification
Purpose & Audience	Domestic plant owners; glanceable moisture, temperature, and hydration alarms.	CONCEPT DEFINED
≥ 3 Navigable Screens	NavigationStack coordinating Dashboard, Plant/Threshold Config, and History views.	DRAFTED IN FLOW
Custom SwiftUI UI	Radial Moisture Gauge, custom view modifiers, and pump state actuators.	COMPONENT DESIGNED
Data Persistence	SwiftData / Core Data SQLite backing for plant models and telemetry historical logs.	TO IMPLEMENT
Emerging Technology	ESP32 IoT Wi-Fi bridge telemetry (or Apple HomeKit / CoreML plant diagnosis).	INTEGRATION SCOPE
Accessibility & Branding	Dynamic Type scaling, high-contrast labels, Dark Mode support, and color-vision parity.	DESIGN COMPLIANT

3. Part B: GreenMind tvOS Living-Room Prototype (10%)

Part B adapts the plant-care ecosystem into an ambient television experience. Designed for a 10-foot viewing distance, the tvOS application displays cards across an expansive grid, enabling home occupants to review plant conditions and execute full CRUD operations using the Apple TV Remote.

PRIMARY DASHBOARD

Collection Grid

Top-level plant cards, total count indicator, and modal trigger.

FOCUSED DETAIL

PlantDetailView

Presents moisture, room temp, care status, and destructive actions.

MODAL SHEETS

Add & Edit Forms

Compact tvOS-calibrated sheets preventing layout clipping.

4. Data Operations & CRUD Interaction Flow

Operation	UI Interaction & State Handling	Verification Criterion
Create	`AddPlantView` sheet appends a new `Plant` instance to the state array.	New card appends instantly to Dashboard grid; opens details cleanly.
Read	Horizontal card focus displaying name, room, moisture %, and care state.	Correct sample values render without truncation across TV resolutions.
Update	`EditPlantView` modifies name and location; isolated by record `UUID`.	Mutates only target plant; updates parent grid and detail views reactively.
Delete	Triggers `confirmationDialog`; removes matching UUID from array on confirm.	Removes card, dismisses detail screen, and restores dashboard focus safely.

5. Technical Architecture & Data Model Implementation

```
      // Swift Model Architecture (tvOS Target) struct Plant: Identifiable { let id: UUID = UUID()
var name: String var location: String var moisture: Int // Sample percentage (0–100%) var
temperature: Int // Sample Celsius reading var needsWater: Bool { moisture < 30 } } // State
Hierarchy: // ContentView (@State var plants: [Plant]) // └── NavigationStack ──> PlantDetailView
(@Binding var plants: [Plant]) // └── .sheet ──────────────> AddPlantView / EditPlantView
```

Prototype Boundaries: The prototype maintains data in-memory via SwiftUI `@State`. Demonstrations utilize verified sample seed entities (*Monstera - Living Room 35%*, *Snake Plant - Bedroom 68%*, *Peace Lily - Kitchen 22%*).

6. Verification Test Suite & Evaluation Matrix

The test suite validates compiler stability, CRUD workflows, remote control focus engines, and data safety boundaries.

ID	Test Scope	Expected Outcome	Execution Status
T01	Compilation Build	Clean build on tvOS simulator with zero duplicate symbols.	VERIFIED (CLEAN)
T02	Plant Creation	Valid submission adds new plant to in-memory array.	VERIFIED
T03	Input Validation	Disallows blank plant names; disables modal save button.	UNDER TEST
T04	UUID Mutability	Editing name/location updates record without creating duplicates.	VERIFIED
T05	Deletion Dialog	Cancelling preserves record; confirming purges target by ID.	VERIFIED
T06	tvOS Remote Focus	Siri Remote highlights active cards with clear visual elevation.	FOCUS AUDITED
T07	App Relaunch Cycle	In-memory data resets cleanly to initial sample seed.	VERIFIED PROTOTYPE

7. Technical Debugging & Layout Calibration

Development of the tvOS target surfaced several critical SwiftUI platform constraints:

1. Invalid Redeclaration of AddPlantView: Occurred when `AddPlantView` was declared in both `ContentView.swift` and its standalone file. Resolved by enforcing modular single-file responsibility.
2. Initializer Argument Mismatch: The plant model expects `location:`, whereas early forms passed `type:`. Corrected to maintain model contract integrity.
3. Sheet Truncation on tvOS: Default iOS sheet dimensions caused oversized buttons and clipped headers on television displays. Calibrated using constrained frame modifiers and compact typography.
4. Identity Stability: Replaced mutable name matching with strict `UUID` identification to prevent data corruption during simultaneous updates.

8. Market Context & Stakeholder Pitch (Part B Requirement)

- **Value Proposition:** Seamless whole-home ambient garden monitoring without requiring access to a mobile handset.
- **Target Market:** Multi-plant smart households, indoor vertical farm owners, and educational urban botanical centers.
- **Revenue Model:** Freemium model with optional cloud sync and multi-sensor hardware hub subscription tier.

9. Demonstration Script & Oral Viva Strategy

1. **Executive Introduction:** Articulate the dual-platform strategy: advanced iOS telemetry paired with tvOS ambient oversight.
2. **tvOS Core Navigation:** Traverse dashboard cards, highlight Siri Remote focus cues, and inspect detail telemetry.
3. **CRUD Demonstration:** Walk through Add Plant sheet submission, UUID-based metadata edit, and modal deletion prompt.
4. **Architectural Justification:** Explain `@State/@Binding` data propagation, `UUID` collision prevention, and the transition plan toward SwiftData persistence.

10. AI-Assisted Development Transparency Disclosure

Per academic integrity mandates, generative AI assistance was utilized exclusively for debugging diagnosis and layout optimization:

Assistance Scope	Action & Guidance Provided	Student Adaptation
Compiler Conflict	Diagnosed duplicate `AddPlantView` symbol collision.	Isolated into dedicated file.
tvOS Screen Calibration	Recommended adjusted spacing and font bounds for 10-foot UI.	Applied compact styles.
Entity Mutability	Identified risks of string-based lookup in mutable collections.	Migrated to UUID indexing.

11. Final Submission Compliance Checklist

- [x] Part A: iOS App Concept & Requirements Defined
- [x] Part B: tvOS Functional Prototype Implemented
- [x] Full CRUD (Create, Read, Update, Delete) Executed
- [x] Siri Remote Focus & Navigation Validated
- [x] GitHub Repositories Structured with Git Commits
- [x] AI Disclosure & Prompt Record Catalogued
- [x] README with Environment & Build Instructions
- [x] Technical Report Completed with Student Metadata

STUDENT ATTESTATION
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